STAT 801A Lab 5: Comparing Two Means

Insert Name

2026-10-06

```
library(ggplot2)
```

Comparing Two Means

Labs 3 & 4 compared one sample mean to a fixed value. This lab compares two independent groups to each other.

The parameter of interest is the difference in population means, $\mu_1 - \mu_2$. The null hypothesis is almost always no difference. The alternative depends on the question.

$$H_0 : \mu_1 = \mu_2 \qquad H_a : \mu_1 \neq \mu_2$$

$$H_0 : \mu_1 = \mu_2 \qquad H_a : \mu_1 > \mu_2$$

$$H_0 : \mu_1 = \mu_2 \qquad H_a : \mu_1 < \mu_2$$

Notation

Symbol	Meaning
μ_1, μ_2	population means
$\bar{y}_1, \bar{y}_2$	sample means
s_1, s_2	sample standard deviations
n_1, n_2	sample sizes

The two-sample t test in this lab assumes:

- The two samples are independent of each other.
- Each population is approximately normal, or the samples are large.
- The two populations have the same standard deviation/variance.

Equal Variances: The Pooled t Test

If both populations have the same standard deviation, each sample is estimating the same thing. Pooling the two samples gives a better estimate than either one alone.

$$s_p^2 = \frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}$$

This is a weighted average of the two sample variances, with the larger sample getting more weight.

The test statistic is

$$t = \frac{\bar{y}_1 - \bar{y}_2}{s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \qquad df = n_1 + n_2 - 2$$

and the confidence interval is

$$(\bar{y}_1 - \bar{y}_2) \pm t_{\alpha/2,\, df} \cdot s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}$$

Example: feed supplement

A feedlot tests whether a feed supplement changes average daily gain (ADG) in steers. Twelve steers get the standard ration and twelve get the ration with the supplement. ADG is measured in pounds per day.

```
steers <- data.frame(
  ration = rep(c("control", "supplement"), each = 12),
  adg = c(3.12, 3.45, 2.98, 3.30, 3.61, 3.05, 3.27, 3.52, 2.89, 3.38, 3.20, 3.41,
          3.48, 3.71, 3.25, 3.66, 3.90, 3.39, 3.58, 3.82, 3.31, 3.55, 3.74, 3.44))

steers
```

```
       ration  adg
1     control 3.12
2     control 3.45
3     control 2.98
4     control 3.30
5     control 3.61
6     control 3.05
7     control 3.27
8     control 3.52
9     control 2.89
10    control 3.38
11    control 3.20
12    control 3.41
13 supplement 3.48
14 supplement 3.71
15 supplement 3.25
16 supplement 3.66
17 supplement 3.90
18 supplement 3.39
19 supplement 3.58
20 supplement 3.82
21 supplement 3.31
22 supplement 3.55
23 supplement 3.74
24 supplement 3.44
```

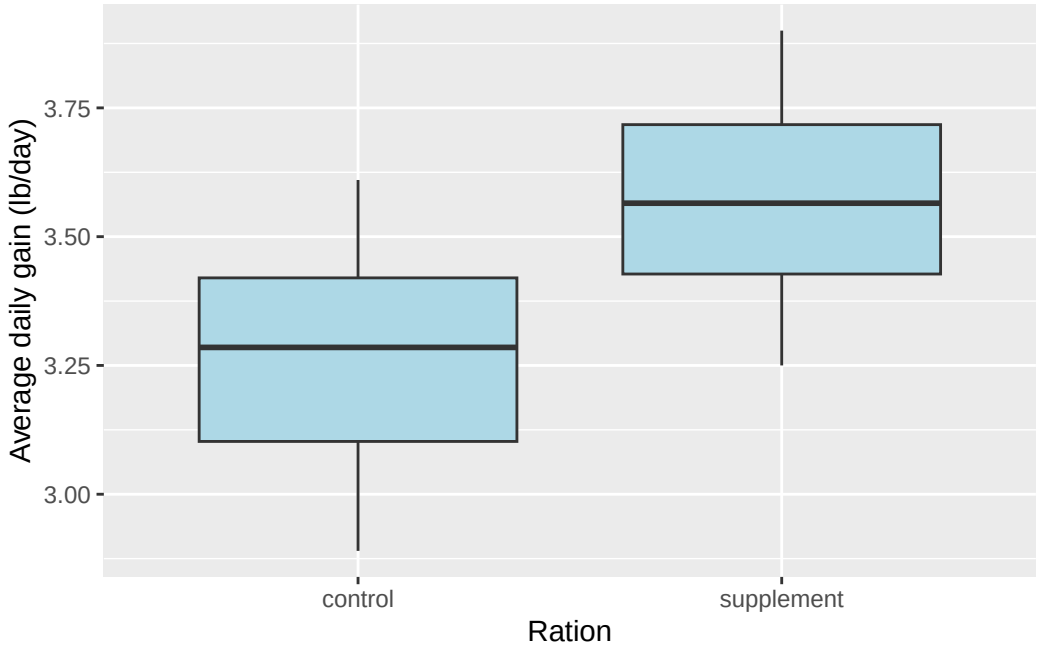
Step 1: State the hypotheses

Group 1 is control and group 2 is supplement. A change in either direction matters, so the test is two-tailed.

$$H_0 : \mu_1 = \mu_2 \qquad H_a : \mu_1 \neq \mu_2$$

Step 2: Look at the data and check the assumptions

```
ggplot(steers, aes(x = ration, y = adg)) +
  geom_boxplot(fill = "lightblue") +
  labs(x = "Ration", y = "Average daily gain (lb/day)")
```




```
control <- steers$adg[steers$ration == "control"]
supplement <- steers$adg[steers$ration == "supplement"]

mean(control)
```

```
[1] 3.265
```

```
mean(supplement)
```

```
[1] 3.569167
```

```
sd(control)
```

```
[1] 0.2230165
```

```
sd(supplement)
```

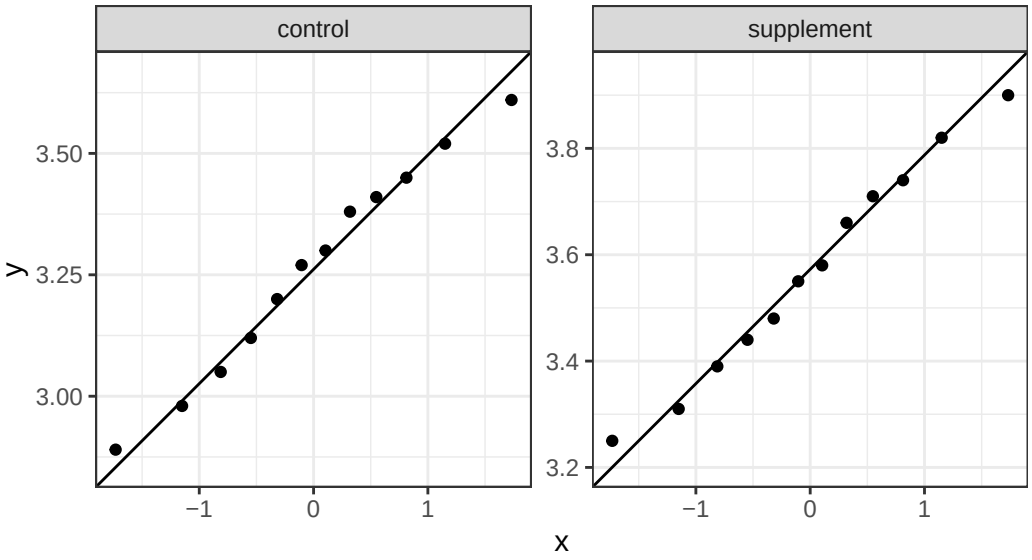
```
[1] 0.2035797
```

The boxplot also works as a rough normality check. Look for a box that is roughly symmetric around the median, whiskers of similar length, and no outlying points. Strong skew and outliers are what cause trouble for a t test.

The boxes are also about the same height, so the equal standard deviation assumption looks reasonable.

We can also check the Q-Q plots for normality. In the plots below, most points follow the line, which indicates that normality appears to hold.

```
ggplot(steers, mapping = aes(sample = adg)) +
  stat_qq() +
  stat_qq_line() +
  theme_bw() +
  theme(aspect.ratio = 1) +
  facet_wrap(~ration, scales = 'free')
```



Step 3: Compute the test statistic

```
n1_steer <- length(control)
n2_steer <- length(supplement)
ybar1_steer <- mean(control)
```

```r
ybar2_steer <- mean(supplement)
s1_steer <- sd(control)
s2_steer <- sd(supplement)

sp_steer <- sqrt(((n1_steer - 1) * s1_steer^2 + (n2_steer - 1) * s2_steer^2) /
                   (n1_steer + n2_steer - 2))

se_steer <- sp_steer * sqrt(1 / n1_steer + 1 / n2_steer)

t_steer <- (ybar1_steer - ybar2_steer) / se_steer

sp_steer
```

```
[1] 0.2135194
```

```r
se_steer
```

```
[1] 0.08716893
```

```r
t_steer
```

```
[1] -3.489393
```

The pooled standard deviation, 0.21, falls between the two sample standard deviations, as a weighted average should.

Step 4: Find the rejection region

```r
df_steer <- n1_steer + n2_steer - 2

qt(0.025, df = df_steer)
```

```
[1] -2.073873
```

```r
qt(0.975, df = df_steer)
```

```
[1] 2.073873
```

We reject if $t < -2.07$ or $t > 2.07$. Our t of -3.49 is below -2.07, so we reject H_0.

Step 5: Compute the p-value

```
2 * pt(abs(t_steer), df = df_steer, lower.tail = FALSE)
```

```
[1] 0.002076252
```

Step 6: Confirm with t.test

```
t.test(adg ~ ration, data = steers, var.equal = TRUE)
```

```
    Two Sample t-test

data:  adg by ration
t = -3.4894, df = 22, p-value = 0.002076
alternative hypothesis: true difference in means between group control and group supplement i
95 percent confidence interval:
 -0.4849440 -0.1233894
sample estimates:
   mean in group control mean in group supplement
                3.265000                 3.569167
```

The argument `var.equal = TRUE` tells R to pool. Without it, R runs a different version of the test that does not assume equal variances, and the output will not match your hand calculation.

Note: R puts the groups in alphabetical order, so it reports control minus supplement.

Step 7: Report a confidence interval

```
margin_steer <- qt(0.975, df = df_steer) * se_steer

(ybar1_steer - ybar2_steer) - margin_steer
```

```
[1] -0.484944
```

```
(ybar1_steer - ybar2_steer) + margin_steer
```

```
[1] -0.1233894
```

This matches the interval from `t.test`. We are 95% confident that the true difference in mean average daily gain (control minus supplement) is between -0.48 and -0.12 lb/day.

Step 8: Write the conclusion

There is strong evidence that the supplement changes average daily gain ($t = -3.49$, $df = 22$, $p = 0.002$). Steers on the supplement gained an estimated 0.30 lb/day more, with a 95% confidence interval of 0.12 to 0.48 lb/day.

Non-Normal Data: The Mann-Whitney U Test

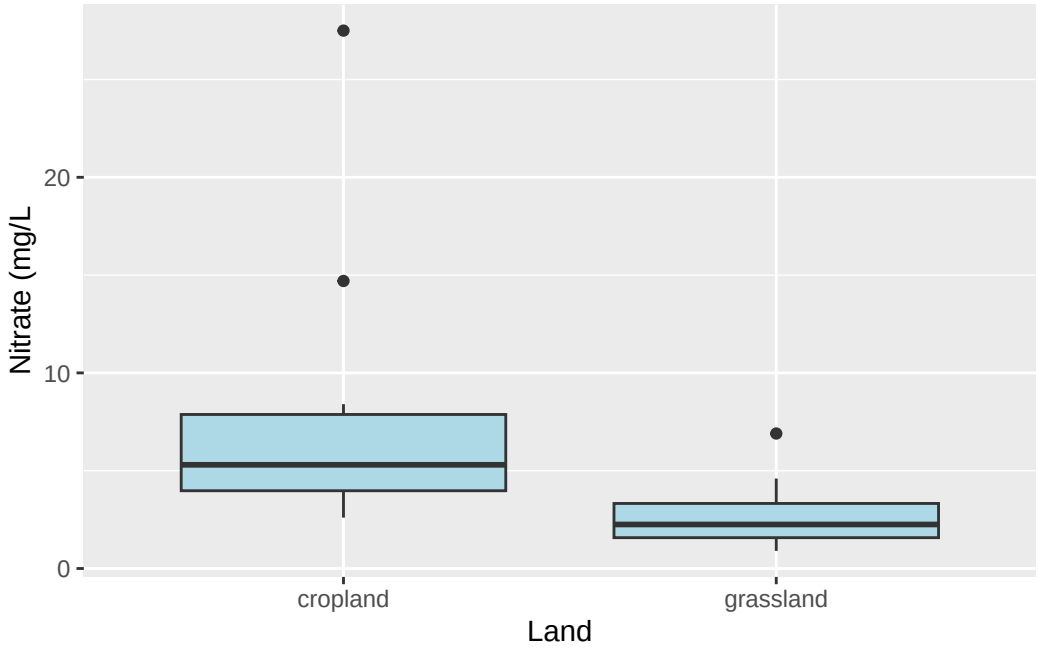
The t test assumes each population is roughly normal. When samples are small and the data are strongly skewed or have outliers, the Mann-Whitney U test is an alternative. Instead of comparing means, it ranks all the observations from both groups together and checks whether one group tends to land in the higher ranks.

In R the test is called `wilcox.test`, after its other name, the Wilcoxon rank-sum test. It uses the same formula setup as `t.test`.

A researcher measures nitrate (mg/L) in 10 private wells near irrigated cropland and 10 wells near grassland.

```
wells <- data.frame(
  land = rep(c("cropland", "grassland"), each = 10),
  nitrate = c(3.1, 5.8, 2.6, 8.4, 4.2, 14.7, 6.3, 3.9, 27.5, 4.8,
              1.2, 2.8, 0.9, 3.5, 1.8, 2.1, 4.6, 1.5, 2.4, 6.9))

ggplot(wells, aes(x = land, y = nitrate)) +
  geom_boxplot(fill = "lightblue") +
  labs(x = "Land", y = "Nitrate (mg/L")
```




```
wilcox.test(nitrate ~ land, data = wells)
```

```
    Wilcoxon rank sum exact test

data:  nitrate by land
W = 86, p-value = 0.005196
alternative hypothesis: true location shift is not equal to 0
```

There is strong evidence that nitrate levels tend to be higher in wells near cropland than in wells near grassland

---

Practice Problems

Where a problem asks for hypotheses, write them in LaTeX.

Problem 1

> A seed company claims its new fungicide increases wheat yield. An agronomist treats 10 plots with the fungicide and leaves 8 plots untreated, then records the yield of each plot in bushels per acre.

Run this chunk to create the data.

```
wheat <- data.frame(
  treatment = rep(c("fungicide", "untreated"), times = c(10, 8)),
  yield = c(61.8, 66.2, 58.4, 63.5, 60.1, 67.9, 62.7, 59.3, 64.8, 61.0,
            59.6, 56.1, 62.8, 57.4, 61.2, 55.0, 61.9, 58.7))
```

(a) Find the sample size, mean, and standard deviation of yield for each treatment.

```
fungicide <- wheat$yield[wheat$treatment == "fungicide"]
untreated <- wheat$yield[wheat$treatment == "untreated"]

length(fungicide)
```

```
[1] 10
```

```
length(untreated)
```

```
[1] 8
```

```
mean(fungicide)
```

```
[1] 62.57
```

```
mean(untreated)
```

```
[1] 59.0875
```

```
sd(fungicide)
```

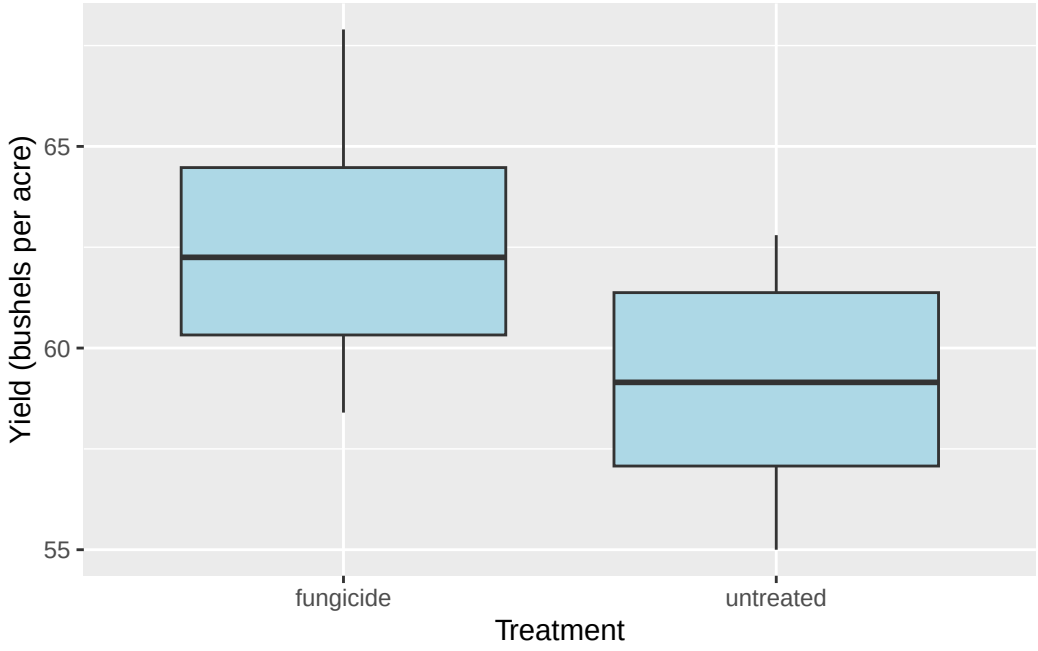
```
[1] 3.067409
```

```r
sd(untreated)
```

```
[1] 2.804302
```

(b) Make side-by-side boxplots of yield by treatment. Do the groups look roughly normal with similar spreads?

```r
ggplot(wheat, aes(x = treatment, y = yield)) +
  geom_boxplot(fill = "lightblue") +
  labs(x = "Treatment", y = "Yield (bushels per acre)")
```



Yes. Both boxes are roughly symmetric with no outliers, and they are about the same height. The standard deviations, 3.07 and 2.80, are also close.

(c) State the hypotheses for testing the company's claim. Let group 1 be fungicide and group 2 be untreated.

$$H_0 : \mu_1 = \mu_2 \qquad H_a : \mu_1 > \mu_2$$

(d) Compute the pooled t statistic by hand.

```r
n1_wheat <- length(fungicide)
n2_wheat <- length(untreated)
ybar1_wheat <- mean(fungicide)
ybar2_wheat <- mean(untreated)
s1_wheat <- sd(fungicide)
s2_wheat <- sd(untreated)

sp_wheat <- sqrt(((n1_wheat - 1) * s1_wheat^2 + (n2_wheat - 1) * s2_wheat^2) /
                   (n1_wheat + n2_wheat - 2))
se_wheat <- sp_wheat * sqrt(1 / n1_wheat + 1 / n2_wheat)
t_wheat <- (ybar1_wheat - ybar2_wheat) / se_wheat

sp_wheat
```

```
[1] 2.955183
```

```r
se_wheat
```

```
[1] 1.401767
```

```r
t_wheat
```

```
[1] 2.484365
```

(e) Find the rejection region at $\alpha = 0.05$ and state your decision.

```r
df_wheat <- n1_wheat + n2_wheat - 2

qt(0.95, df = df_wheat)
```

```
[1] 1.745884
```

This is upper-tailed, so all 5% goes in the upper tail. We reject if $t > 1.75$. Our t of 2.48 is above 1.75, so we reject H_0.

(f) Compute the p-value.

```r
pt(t_wheat, df = df_wheat, lower.tail = FALSE)
```

```
[1] 0.01221506
```

The p-value is 0.012.

(g) Confirm your work with `t.test`.

```r
t.test(yield ~ treatment, data = wheat, var.equal = TRUE, alternative = "greater")
```

```
    Two Sample t-test

data:  yield by treatment
t = 2.4844, df = 16, p-value = 0.01222
alternative hypothesis: true difference in means between group fungicide and group untreated
95 percent confidence interval:
 1.035179      Inf
sample estimates:
mean in group fungicide mean in group untreated
                62.5700                 59.0875
```

Fungicide comes first alphabetically, so R tests fungicide minus untreated, which matches our hypotheses.

(h) Construct a 95% confidence interval for $\mu_1 - \mu_2$ by hand and interpret it.

```r
margin_wheat <- qt(0.975, df = df_wheat) * se_wheat

(ybar1_wheat - ybar2_wheat) - margin_wheat
```

```
[1] 0.5108875
```

```r
(ybar1_wheat - ybar2_wheat) + margin_wheat
```

```
[1] 6.454112
```

We are 95% confident that the true difference in mean yield (fungicide minus untreated) is between 0.51 and 6.45 bushels per acre.

(i) Write a conclusion in context.

There is evidence that the fungicide increases mean wheat yield ($t = 2.48$, $df = 16$, $p = 0.012$). Treated plots averaged 3.5 bushels per acre more than untreated plots, with a 95% confidence interval of 0.5 to 6.5 bushels per acre.

Problem 2

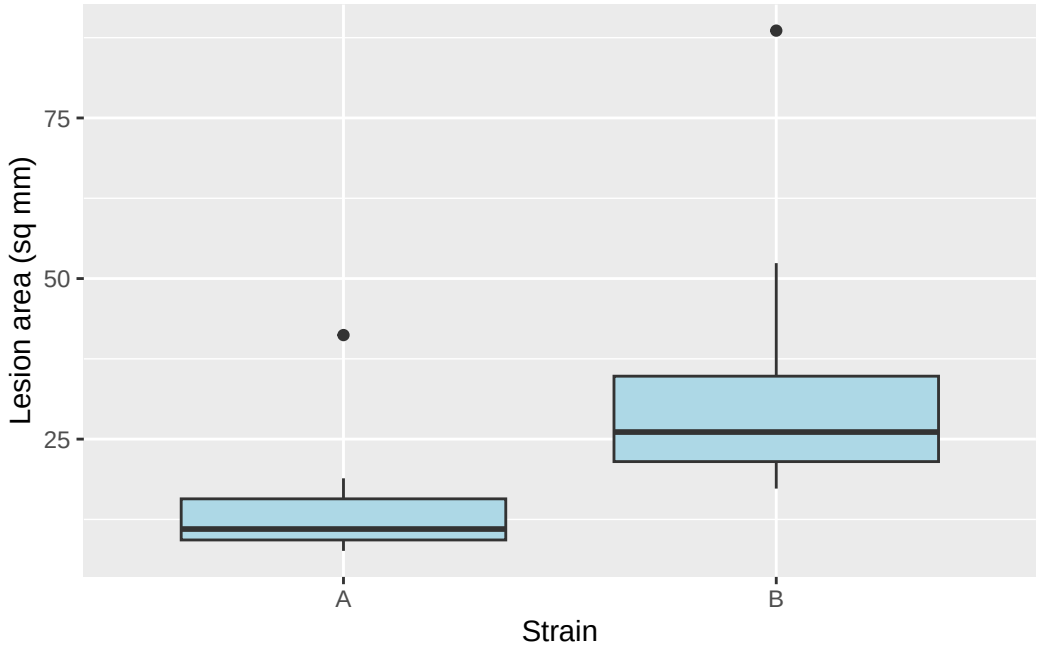
> A plant pathologist inoculates leaves with one of two fungal strains, A or B, and measures the lesion area (mm^2) on each leaf after one week. Nine leaves receive each strain.

Run this chunk to create the data.

```
lesions <- data.frame(
  strain = rep(c("A", "B"), each = 9),
  area = c(12.4, 8.1, 15.7, 9.3, 41.2, 11.0, 7.6, 18.9, 10.2,
           21.5, 34.8, 17.3, 88.6, 26.1, 19.7, 52.4, 23.9, 30.2))
```

(a) Make side-by-side boxplots of lesion area by strain. Why might a t test be a poor choice here?

```
ggplot(lesions, aes(x = strain, y = area)) +
  geom_boxplot(fill = "lightblue") +
  labs(x = "Strain", y = "Lesion area (sq mm)")
```



Both groups are right-skewed: each median sits near the bottom of its box, and each group has a high outlier (41.2 for strain A, 88.6 for strain B). With only nine leaves per strain, normality is not a safe assumption.

(b) Run the Mann-Whitney test. Is there evidence that the two strains differ?

```
wilcox.test(area ~ strain, data = lesions)
```

```
    Wilcoxon rank sum exact test

data:  area by strain
W = 8, p-value = 0.002756
alternative hypothesis: true location shift is not equal to 0
```

Yes. The p-value is 0.003, so there is strong evidence that lesion areas differ between the two strains.

Final Step

Render your document. Click the Render button at the top of the editor, or press Ctrl + Shift + K.

If the render fails, read the error message. It names the chunk that caused the problem. The most common causes are a missing closing parenthesis, or an object used before it was created.